

Dave's Daily

AI • Bayesian optimization • US equities

tldr

1. Trump put his intelligence chief, Jay Clayton, in charge of a new “Super Intelligence Force,” five days after six AI company leaders signed a safety accord that names no auditor. In a new poll, 73% of Americans worry AI could threaten human survival.
2. AI’s profits are landing in memory chips. Micron’s quarterly revenue rose 379% in a year, to \$54.2 billion, at an 87% gross margin. Meanwhile the equal-weighted S&P 500 trailed the index by 4.6 points in September.
3. Citadel is hiring AI-lab researchers as “managers of machines.” Its quant chief says short-horizon trading edges are now learned and copied fast, and high-frequency firms are drifting toward longer holding periods.

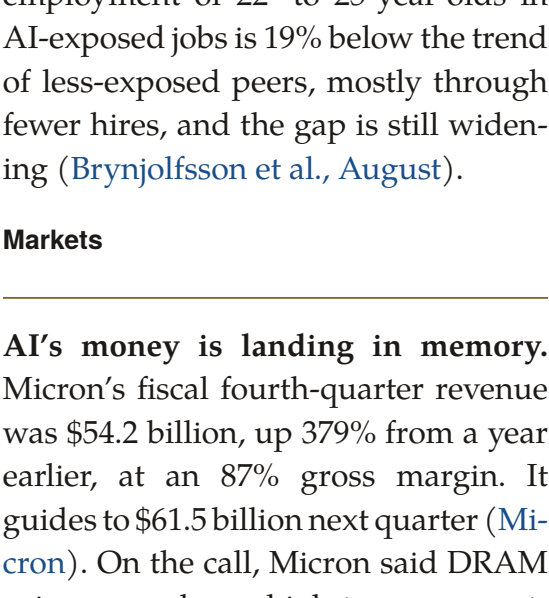
AI

A “Super Intelligence Force” with no stated powers. On Sunday President Trump named Director of National Intelligence Jay Clayton to lead a new task force, joined by FTC chair Andrew Ferguson, OPM director Scott Kuper, and Pentagon technology chief Emil Michael. It reports to Trump and chief of staff Susie Wiles ([CBS](#); [NPR](#); [Washington Post](#)). Its stated job is to “coordinate” the government’s dealings with consumers, religious groups, infrastructure providers, and AI companies. It follows the 29 September accord in which six AI company leaders promised “four layers of controls and audits.” That accord names no auditor and sets no deadline or penalty. Hypothesis: pairing a spy chief with the consumer-protection regulator suggests the White House sees AI oversight as security plus consumer protection, not licensing.

Benchmarks versus work. Google began rolling out Gemini 4 Argon on 30 September, first to vetted cyber defenders, citing leading benchmark scores. Insiders told [Bloomberg](#) that it does worse on real coding work, especially front-end design, and two blamed “benchmaxxing,” tuning for the test rather than the job. Google calls that inaccurate ([9to5Google](#)).

TPUs in orbit, in 15-minute bursts. Google’s first Project Suncatcher satellite, carrying four Trillium TPUs, reached orbit on 1 October and is working ([Google](#)). The hard limit is heat: in vacuum the radiators can shed full-load heat for only about 15 minutes before a cool-down ([TechTimes](#)).

The public wants brakes



People: \$10,000 a household is not enough. NorthPoint Development offered each of about 4,500 households in Hazle Township, Pennsylvania, \$10,000 if its 1,300-acre, 300-megawatt data-center campus is approved ([WSJ](#); [Bisnow](#)). That is over 15% of the town’s median household income. Many residents call it a bribe. The township rejected the plan on zoning grounds and the developer is appealing ([FOX56](#)). The chart’s 11-to-65 split shows Hazle is not unusual.

Jobs: the entry door narrows. Stanford’s updated ADP payroll study finds no economy-wide job loss. But employment of 22- to 25-year-olds in AI-exposed jobs is 19% below the trend of less-exposed peers, mostly through fewer hires, and the gap is still widening ([Brynjolfsson et al.](#), [August](#)).

Markets

AI’s money is landing in memory. Micron’s fiscal fourth-quarter revenue was \$54.2 billion, up 379% from a year earlier, at an 87% gross margin. It guides to \$61.5 billion next quarter ([Micron](#)). On the call, Micron said DRAM prices rose by a high-teens percentage in one quarter and NAND prices by about 30%. It has 26 take-or-pay customer agreements, with remaining performance obligations of about \$150 billion ([transcript](#)). Hypothesis: long contracts at floor prices make memory, a famously cyclical business, look more like a utility. Every phone and PC maker is paying the bill.

A narrow market. In September the equal-weighted S&P 500 fund (RSP) trailed the cap-weighted one (SPY) by 4.6 percentage points. Bespoke finds only two worse months since 2003, March 2020 and March 2023 ([CNBC](#)). CFRA’s Sam Stovall counts only 12% of the 153 sub-industries in the S&P 1500 trading above both their 50- and 200-day averages. The index is near its high because a few AI names carry it.

Citadel: edges at short horizons decay fastest. Citadel’s quant group, about 180 people, plans “healthy” double-digit growth and is recruiting from labs such as Google DeepMind ([Bloomberg via Moneycontrol](#); [Hedgeweek](#)). Quant chief Navneet Arora says AI lowers the premium on coding and raises it on idea generation: researchers become “managers of machines.” In short-horizon trading, plentiful price and volume data let machines learn fast, and slim edges are copied fast. Longer-horizon signals rest on thinner data and may last longer. High-frequency firms are moving that way, while Citadel moves toward shorter horizons.

Key numbers (this edition)

Item	Figure
Micron revenue, year over year	+379%
Micron gross margin, FQ4	87.0%
RSP minus SPY, September	−4.6 pts
Young AI-exposed workers vs. trend	−19%
Hazle offer, per household	\$10,000

Bayesian optimization

Scaling: why acquisition gradients vanish. Saksham Kiroriwal, Julius Pfrommer, and Jürgen Beyerer of Fraunhofer IOSB and KIT pull the Fisher information metric back through the surrogate onto the input space ([arXiv:2609.31107](#), 25 Sep). The result bounds the acquisition gradient by two factors: how sensitive the acquisition is to the predictive distribution, and whether moving in input space changes that distribution at all. In high dimensions the second factor collapses far from data. The analysis explains dimension-scaled lengthscale priors and RAASP with one mechanism. Their method, FITR, shapes TuRBO’s trust region with local Fisher weights. It beats TuRBO-REI on several benchmarks and works with kernels that have no lengthscales.

Data efficiency: many fidelities, with constraints. Lucas Palazzolo, Mickaël Binois, and Laetitia Giralaldi of Inria extend scalable constrained BO to any number of cheap, approximate fidelities ([arXiv:2609.29941](#), 24 Sep). They say no earlier method combined high dimension, black-box constraints, many fidelities, and non-nested samples. On tests up to 100 dimensions it generally converges faster than single-fidelity SCBO.

Connections to AI: learning the geometry. Edwin Ketabati Augustinsson and Robert Bridges let GP-UCB choose input warps, such as log scaling, from a finite library by any rule, and keep its no-regret guarantee ([arXiv:2609.02993](#)). On Fashion-MNIST hyperparameter tuning it recovers much of the gain of hand-set log scales. A 20-dimensional multi-agent design study shows it is feasible under costly, noisy evaluations. Hypothesis: both papers suggest high-dimensional BO is now less about new kernels and more about fixing the geometry the optimizer sees.